



3 A computer has a central processing unit (CPU), memory and secondary storage.

(a) The computer has a Von Neumann architecture.

Circle **three** registers that are found in the Von Neumann architecture.

accumulator	arithmetic register	binary counter
binary register	control register	current data register
current instruction register	data counter	instruction counter
logical register	program counter	storage register

[3]

(b) State the purpose of the CPU in the computer.

.....
 [1]

(c) Computer A has a 2.1 GHz quad-core processor.

Computer B has a 2.5 GHz dual-core processor.

Computer C has a 5.2 GHz single-core processor.

(i) Identify which computer is most likely to execute more instructions simultaneously.

Justify your answer.

Computer

Justification

 [3]

(ii) Explain why Computer C can execute more instructions per second than Computer B.

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 [2]

